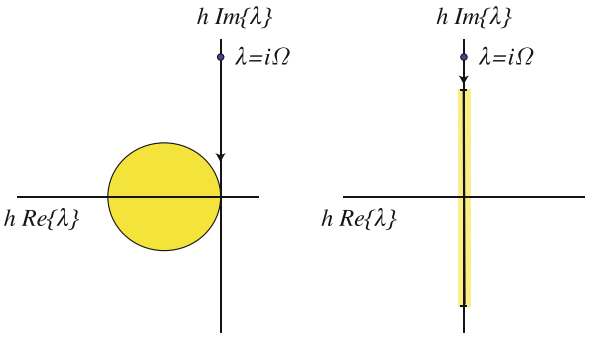


Fig. 4.1 Stability regions for Euler's method (*left*) and Symplectic Euler/Verlet (*right*). When a harmonic oscillator is treated using these methods, the origin is unstable for Euler's method, regardless of stepsize—this means that there is no choice of scaling h which will allow us to ensure that $|1 + h\lambda| \leq 1$. On the other hand, the Verlet method has an interval of stability on the imaginary axis, and it is always possible to find a value of h which guarantees that $|h\Omega| \leq 2$

general nonlinear systems. It is possible for the linearized system to be stable and for the equilibrium point nonetheless to be unstable, since, with Hamiltonian systems, we typically encounter the purely imaginary eigenvalue case, but the linearized analysis is nonetheless helpful. For example, applying the Verlet method to the harmonic oscillator with frequency Ω we find that the origin is stable (and the numerical solution stays bounded for all time) provided $h\Omega \leq 2$ (the same condition as for stability of Symplectic Euler). The situation with regard to Euler and Verlet (or Symplectic Euler) methods for the harmonic oscillator is illustrated in Fig. 4.1.

In the context of molecular dynamics involving chemical bonds, the linear stability condition $h\Omega \leq 2$ is often quoted as the stepsize limiting condition where Ω is assumed to the frequency of the fastest (highest energy) bond stretch, Ω_F, present in the molecular dynamics model. This is realistic because the bond stretches are the highest frequency components in general in these models, and the potential energy function used for the bond stretch is typically quadratic in the separation distance:

$$U_{\text{bond}}(\boldsymbol{q}_i, \boldsymbol{q}_j) = \frac{\kappa}{2} \left(\|\boldsymbol{q}_i - \boldsymbol{q}_j\| - l_{ij} \right)^2 ,$$

for positive parameters κ and l_{ij}. It should be noted that even this term leads to a nonlinear differential equation system and so the relation to the harmonic model should be understood as being of a heuristic nature; it is only through numerical experiment that the correspondence is ultimately established.

More generally, even without the presence of bonds, the "stepsize threshold" in the simulation of molecular models by an explicit method like Verlet arises because of the shape of the energy surface. The understanding of this for a nonlinear system like the Lennard-Jones cluster requires examining the linearization of the problem near equilibrium points and even then the discussion is always quite technical